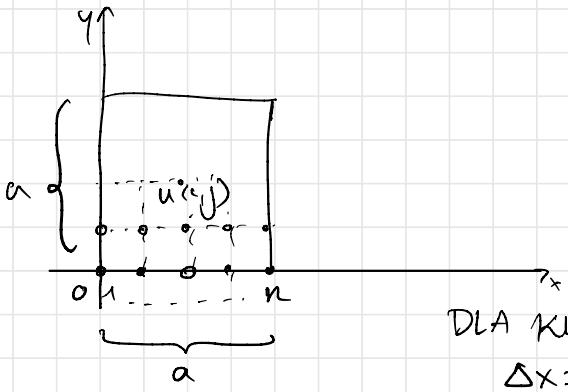
Metody numeryczne:

I. Metoda relaksacyjna



DLA KWADRATU

$$\Delta x = \frac{a}{N}$$

$$\Delta y = \frac{a}{N} = \Delta x$$

$$\frac{\partial v}{\partial x} = \lim_{\Delta x \to 0} \frac{v(x+\Delta x, y) - v(x,y)}{\Delta x}$$

$$\frac{\partial u}{\partial x} \approx \frac{u(x+\Delta x, y) - v(x,y)}{\Delta x}$$

$$h = \Delta x = \Delta x$$

$$x_i = \Delta x \cdot i$$

$$y_j = \Delta x \cdot j = \Delta y \cdot j$$

$$u(x,y) \longrightarrow u_{i,j}$$

$$u(x+h, y) \longrightarrow u_{i+1,j}$$

$$u(x, y+h) \longrightarrow u_{i,j+1}$$

$$\frac{\partial u}{\partial x}(x_i, y_j) \approx \frac{u_{i+1,j} - u_{i,j}}{h}$$

$$\frac{\partial v}{\partial y}(x_i, y_j) \approx \frac{u_{i,j+1} - u_{i,j}}{h}$$

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{\frac{\partial u}{\partial x}(x_{i+1}, y_j) - \frac{\partial u}{\partial x}(i,j)}{h}$$

$$= \frac{\frac{u(x_{i+2}, y_j) - u(x_{i+1}, y_i)}{h} - \frac{u(x_{i+1}, y_i) - u(x_i, y_j)}{h}}{h}$$

$$= \frac{u(x_{i+2}, y_j) - 2u(x_{i+1}, y_i) + u(x_i, y_j)}{h^2}$$

$i \to i-1$

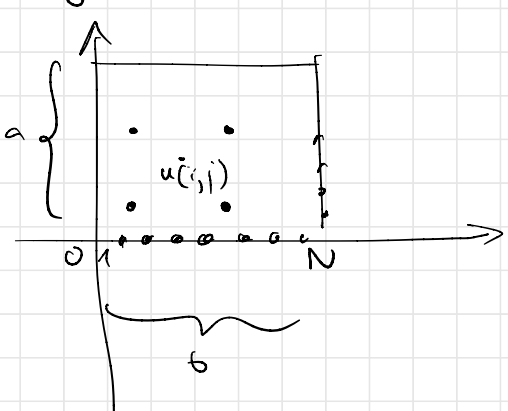
$$\frac{\partial^2 u}{\partial x^2}(x_i, y_j) \approx \frac{u(x_{i+1}, y_j) - 2u(x_i, y_j) + u(x_{i-1}, j)}{h^2}$$

$$\frac{\partial^2 u}{\partial y^2}(x_i, y_j) = \frac{u(x_i, y_{j+1}) - 2u(x_i, y_j) + u(x_i, y_{j-1})}{h^2}$$

$+$

$$\frac{1}{h^2}\left(u(x_{i+1}, y_j) + u(x_{i-1}, y_j) + u(x_i, j+1) + u(x_i, y_{j-1}) - 4u(x_i, y_i)\right) = 0 \quad | \cdot h^2$$

$$u(x_i, y_i) = \frac{u(x_{i+1}, y_j) + u(x_{i-1}, y_j) + u(x_i, y_{j+1}) + u(x_i, y_{j-1})}{4}$$



$u(0,j) = f(j)$

$u(N,j) = f(j)$

$u(i,0) = f(i)$

$u(i,N) = f(i)$